Requirements Gathering

This document covers the four Requirements Gathering deliverables from the project guidelines: stakeholder analysis, user stories and use cases, functional requirements, and non-functional requirements. Priorities use MoSCoW (Must, Should, Could). Targets match the KPIs in the Planning document.

1. Stakeholder Analysis

Stakeholder	Role	Needs from the system	Influence	Interest
Data engineers (the team)	Build, run and maintain the pipeline	Clear setup, monitoring metrics, replay of raw data, simple debugging	High	High
Data analysts	Query the clean and aggregated data	Trusted, deduplicated data in a relational schema, documented columns	Medium	High
Dashboard consumers (operations or facility managers)	Watch device and site status	Live readings, device health, clear data-quality indicators	Medium	High
Device and field technicians	Respond to faults	Clear view of offline devices and out-of-range readings	Low	Medium
Lecturers and assessors	Evaluate the project	Complete documents, working demo, measured KPIs, clean GitHub repository	High	High
Platform and DevOps support (the team)	Deploy and operate the stack	One-command setup, pinned versions, health checks	Medium	Medium

2. User Stories and Use Cases

2.1 User stories

ID	As a	I want	So that	Related FR	Priority
US-01	Data engineer	to start the whole pipeline with one command	anyone can run it without manual setup	FR-21	Must
US-02	Data engineer	to simulate many devices with configurable rate and fault injection	I can test normal and bad-data conditions	FR-01, FR-02	Must
US-03	Data engineer	invalid events sent to a dead-letter store with the reason	bad data is never lost or silently dropped	FR-08, FR-09	Must
US-04	Data engineer	duplicate events removed	counts and averages are correct	FR-10	Must
US-05	Data engineer	late events handled by the approved watermark policy (2-minute start value)	windows stay correct without unbounded state	FR-11	Must
US-06	Data engineer	raw events kept unchanged	I can replay and reprocess after a bug fix	FR-12	Should
US-07	Data analyst	clean readings in a relational table with keys	I can query them with SQL	FR-13, FR-14	Must
US-08	Data analyst	per-minute and per-site aggregates	I can analyze trends quickly	FR-15	Must
US-09	Dashboard consumer	to see device readings within seconds of arrival	I can react to what is happening now	FR-16	Must
US-10	Dashboard consumer	to see which devices are offline or silent	I know where data is missing	FR-17	Should
US-11	Dashboard consumer	a data-quality panel showing rejected and late events	I know how far to trust the numbers	FR-18	Should
US-12	Operations manager	threshold alerts shown on the dashboard	I notice dangerous readings fast	FR-19	Could
US-13	Data engineer	latency, throughput and lag metrics	I can prove the KPIs are met	FR-20	Must

ID	As a	I want	So that	Related FR	Priority
US-14	Assessor	tests, documents and a README in GitHub	I can verify the work	FR-22, FR-23, FR-28	Must
US-15	Data engineer	data-quality rules defined as Great Expectations suites with a daily report	quality is checked repeatably and visible to everyone	FR-24	Must
US-16	Data engineer	Airflow DAGs for stream health, daily quality report, lake compaction, gold backfill and warehouse maintenance	routine operations run on schedule and on demand	FR-25	Must
US-17	Data engineer	failed writes retried without creating duplicates	no valid event is lost or stored twice	FR-26, FR-27	Must

2.2 Use cases

The use case diagram is part of the Design stage. These are the use cases it will show.

ID	Use case	Actor	Summary
UC-01	Generate sensor events	Simulator (system actor), data engineer	Simulator creates events at a configured rate, with optional faults, and publishes to Kafka
UC-02	Validate and clean events	Streaming job	Parses events, checks schema and ranges, removes duplicates, routes bad events to the dead-letter topic
UC-03	Aggregate readings	Streaming job	Computes windowed aggregates per device and per site using event time
UC-04	Store and serve data	Streaming job, SQL Server	Writes raw, clean and aggregated layers and updates serving tables
UC-05	View live dashboard	Dashboard consumer	Opens the dashboard and sees live readings, device health and quality indicators
UC-06	Query historical data	Data analyst	Runs SQL on the clean and aggregate tables

ID	Use case	Actor	Summary
UC-07	Monitor pipeline health	Data engineer	Reads latency, throughput, lag and error metrics
UC-08	Inspect and replay rejected or raw data	Data engineer	Reviews dead-letter records and replays raw data after a fix
UC-09	Run scheduled operations	Airflow, data engineer	Airflow runs stream health checks, the daily quality report, lake compaction, gold backfill and warehouse maintenance on schedule or on demand

UC-02 in detail (example)

Field	Description
Actor	Streaming job
Precondition	Raw events are available on the Kafka topic
Trigger	A micro-batch of events arrives
Main flow	1. Parse the event. 2. Check required fields, types and value ranges. 3. Drop duplicates by event ID within the watermark. 4. Write valid unique events to the clean layer and the serving table. 5. Verify the required writes
Alternate flow: invalid	Parsing, schema or range check fails: send the original payload and the reason to the dead-letter topic, then continue
Alternate flow: duplicate	The event is suppressed from the clean and serving layers and counted
Alternate flow: late	Events later than the 2-minute watermark go to the late-events Parquet path and are corrected by the gold_backfill DAG
Alternate flow: infrastructure failure	The write is retried with idempotent writes; the event is not counted as persisted until the required writes are verified
Postcondition	Every event is accounted for as stored, rejected to the dead-letter store, suppressed as a duplicate,

Field	Description
	handled as late under the approved policy, or still being retried

3. Functional Requirements

ID	Area	Requirement	Priority
FR-01	Ingestion	The simulator shall generate events for a configurable number of devices and sites at a configurable rate	Must
FR-02	Ingestion	The simulator shall inject late, duplicate, malformed and out-of-range events at configurable percentages	Must
FR-03	Ingestion	Each event shall carry a unique event ID, device ID, site ID, sensor type, value, unit and event timestamp	Must
FR-04	Ingestion	Events shall be published to a Kafka topic partitioned by device ID	Must
FR-05	Ingestion	A separate Kafka dead-letter topic shall receive rejected events	Must
FR-06	Processing	The streaming job shall consume the raw topic and parse events against a defined schema	Must
FR-07	Processing	The job shall enrich events with device and site reference data	Should
FR-08	Processing	The job shall validate required fields, types and value ranges per sensor type	Must
FR-09	Processing	A consumer shall persist records from the dead-letter topic to durable dead-letter storage with the failure reason and original payload	Must
FR-10	Processing	The job shall remove duplicate events by event ID	Must
FR-11	Processing	The job shall use event-time windows with a watermark to handle late events; the watermark starts at 2 minutes and changes only through a pull request with test evidence, and later events go to the late-events Parquet path and are corrected by gold_backfill	Must
FR-12	Storage	Raw events shall be stored unchanged in the raw Parquet layer; the Kafka raw topic keeps 7 days and	Should

ID	Area	Requirement	Priority
		the raw Parquet layer is kept for the whole project, so replay after 7 days uses Parquet	
FR-13	Storage	Validated events shall be stored in the clean Parquet layer and in a SQL Server Developer table	Must
FR-14	Storage	The serving schema shall use primary and foreign keys for devices, sites and readings	Must
FR-15	Storage	Windowed aggregates (average, minimum, maximum, count) shall be stored per device and per site	Must
FR-16	Serving	The Grafana dashboard shall show live readings and aggregates, refreshing at an interval no longer than the freshness target	Must
FR-17	Serving	The dashboard shall show device status (active or silent) based on last event time	Should
FR-18	Serving	The dashboard shall show data-quality indicators: rejected, duplicate and late event counts	Should
FR-19	Alerting	The dashboard shall highlight readings that cross configured thresholds	Could
FR-20	Operations	The pipeline shall record event, ingest, processed and served timestamps and expose latency, throughput and consumer lag	Must
FR-21	Operations	The full stack shall start with one Docker Compose command and stop cleanly	Must
FR-22	Quality	The project shall include pytest unit tests and data quality checks	Must
FR-23	Documentation	The repository shall include a README with installation, configuration and execution steps	Must
FR-24	Quality	Data quality rules shall be defined as Great Expectations expectation suites that validate the stored clean Parquet and SQL Server tables, run on a schedule and on demand, with stored validation results and a daily report	Must
FR-25	Orchestration	Airflow shall run five DAGs: stream_health_check, dq_daily_report, lake_compaction, gold_backfill and warehouse_maintenance	Must

ID	Area	Requirement	Priority
FR-26	Reliability	Failed writes shall be retried, writes shall be idempotent so retries create no duplicates, and an event counts as persisted only after the required writes are verified	Must
FR-27	Operations	A reconciliation check shall compare produced, stored, rejected and duplicate-suppressed event counts	Should
FR-28	Documentation	All project files (documents, diagrams, source code, Airflow DAGs, Great Expectations suites, tests, reports, presentation, infographic) shall be uploaded to the GitHub repository using the folder structure in the Planning document, section 2.3	Must

Event data requirements

Field	Type	Rule
event_id	string	Required; UUID created by the producer, globally unique, reused when a send is retried
device_id	string	Required; must exist in the device registry
site_id	string	Required; must match the site registered for the device, otherwise dead-letter
sensor_type	string	One of temperature, humidity, co2, power; unknown types go to the dead-letter queue
value	double	Required, not null; temperature -40 to 85 °C, humidity 0 to 100 %, CO2 0 to 5,000 ppm, power 0 to 5,000 W
unit	string	Matches the sensor type (°C, %, ppm, W)
event_ts	timestamp	Required; ISO 8601 in UTC with Z; rejected if more than 5 s in the future

Other rules: missing or null required fields go to the dead-letter queue, and extra fields are dropped and counted. Planned simulation size: 500 devices sending one reading per second, which is about 500 events per second, with peak tests at 1,000 events per second.

Acceptance evidence

Each Must requirement needs a testable outcome. These are suggested evidence standards; the project guidelines and lecturer instructions remain authoritative. Test IDs are assigned in the test plan.

Requirements	Suggested acceptance evidence	Test ID
FR-01 to FR-06	Simulator configuration, Kafka topic configuration (raw and dead-letter) and a measured production-rate test	To be assigned
FR-08, FR-09	Test fixtures showing invalid records captured with correct rejection reasons	To be assigned
FR-10	SQL and Parquet checks showing no duplicate event IDs in the clean and serving layers	To be assigned
FR-11	Tests for events inside and outside the 2-minute watermark	To be assigned
FR-13 to FR-15	Database schema, successful writes and verified aggregate calculations	To be assigned
FR-16 to FR-20	Dashboard screenshots plus timestamped latency, freshness and throughput measurements	To be assigned
FR-21	Clean-start test using the documented Docker Compose command	To be assigned
FR-22, FR-24	Automated test results and the Great Expectations validation report	To be assigned
FR-23, FR-28	README walkthrough completed by a team member other than its author; repository checked against the folder structure with every deliverable linked	To be assigned
FR-25	Each of the five DAGs run on schedule and on demand with logged results	To be assigned
FR-26, FR-27	Forced write failure and service restart showing retries without duplicates, and a passing reconciliation	To be assigned

4. Non-Functional Requirements

ID	Category	Requirement	Target	Priority
NFR-01	Performance	Processing latency from Kafka ingest time to the serving layer, with event-time delay reported separately	p95 of 10 s or less	Must
NFR-02	Performance	Sustained throughput on a laptop	500 events/s, with a 1,000 events/s peak test	Must
NFR-03	Performance	Dashboard data age	15 s or less	Must
NFR-04	Reliability	No data loss between broker and storage	100% of valid, unique events stored (intentional duplicate suppression excluded)	Must
NFR-05	Reliability	Pipeline availability and recovery	99% over a 2-hour soak; restart recovery within 60 s	Should
NFR-06	Reliability	Failed records handled without stopping the job	Zero job crashes from bad records	Must
NFR-07	Scalability	Throughput grows by adding Kafka partitions and Spark resources	Documented scaling path; tested with at least two partition counts, recording partition counts, resources, test duration and throughput and lag results	Should
NFR-08	Data quality	Duplicates in clean and serving layers	0	Must
NFR-09	Data quality	Invalid injected events captured in the dead-letter store	100% with a reason	Must

ID	Category	Requirement	Target	Priority
NFR-10	Security	No credentials in the repository; secrets in environment files not committed; services reachable only on the local network	Checked in code review	Must
NFR-11	Maintainability	Modular, commented code with consistent naming and PR review	Every merge through a reviewed PR	Should
NFR-12	Portability	Runs on Windows, macOS or Linux with Docker	Setup tested on at least two operating systems	Should
NFR-13	Usability	A new user can start the pipeline and read the dashboard from the manual	Setup in 15 minutes or less	Should
NFR-14	Observability	Pipeline health visible without reading logs	Metrics panel on the dashboard	Should

5. Traceability

Objective (Planning document)	Requirements
O1 Ingest at 500 events/s	FR-01 to FR-05, NFR-02
O2 Streaming validation, deduplication, windows	FR-06 to FR-11, NFR-06, NFR-08
O3 Dead-letter handling	FR-05, FR-09, FR-24, NFR-09
O4 Layered storage and relational serving	FR-12 to FR-15, FR-26
O5 Live dashboard	FR-16 to FR-19, NFR-03
O6 Measured KPIs	FR-20, NFR-01 to NFR-05
O7 Full guideline deliverables	FR-21 to FR-23, FR-28, NFR-10 to NFR-14
O8 Orchestration with Airflow	FR-25, FR-27